

Research Unplugged: Introduction to R & RStudio

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What is R?

- ▶ Open-source language designed for **statistical computing, analysis and open science**.
- ▶ Extensive Package Ecosystem: features over 20,000 packages (via CRAN) that allow users to perform several different statistical models and visualisations.
- ▶ Renowned for its ability to produce high quality graphics and interactive web apps

What is RStudio?

- ▶ It is an integrated development environment (IDE)
- ▶ Built for Data Science
- ▶ Workspace for Scripting, academic Manuscripts, Presentations, and more...

Alternatives?

- ▶ What about alternative programs such Python, SPSS, SAS, Stata, Minitab, Matlab, Maple, Mathematica, Tableau, Excel, SQL and many others) come down to 3 inter-related factors:
 - ▶ It is immensely powerful
 - ▶ Open source
 - ▶ Very and increasingly widely used
- ▶ Positron (IDE)
 - ▶ New upcoming IDE
 - ▶ Worth it?

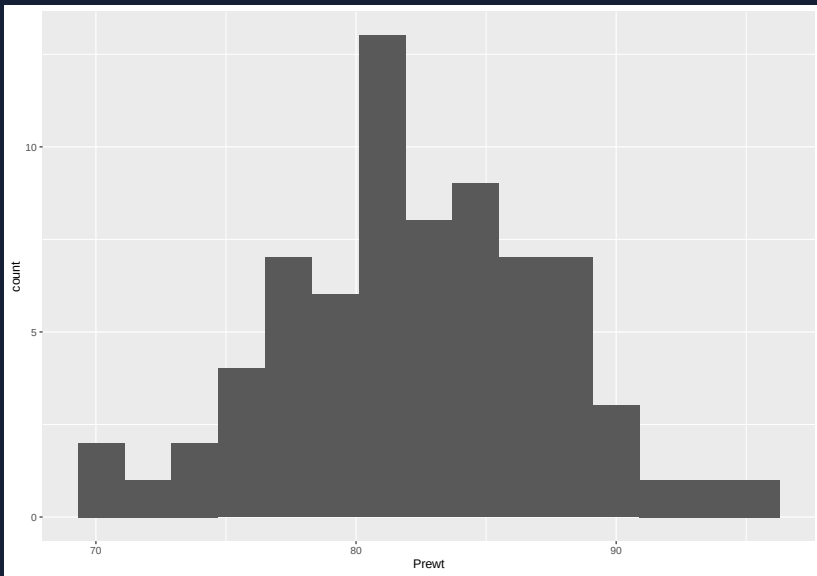
Essential Concepts

- ▶ Data Wrangling
- ▶ Data Visualisation
- ▶ Quarto
- ▶ Statistics

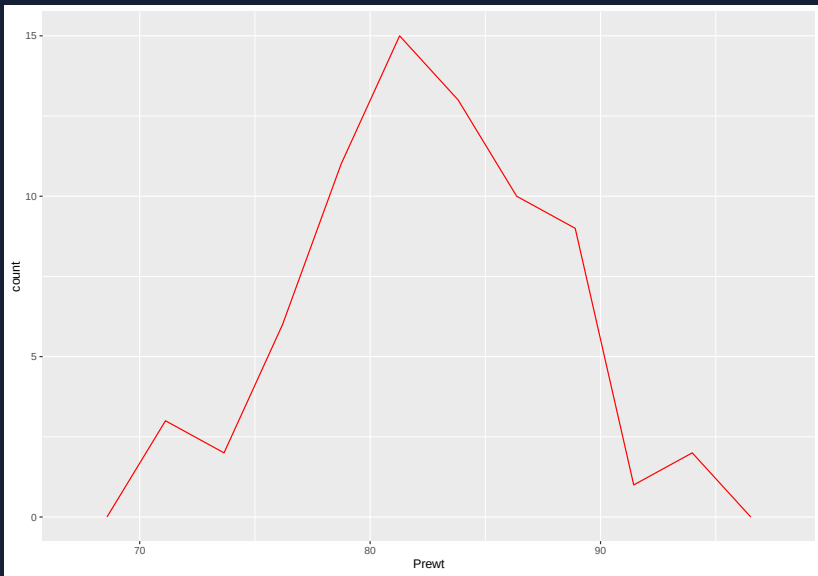
What are we going to learn today

- ▶ *Guided tour of RStudio*
 - ▶ Calling R packages, functions, variables etc.
- ▶ *First Steps in R*
- ▶ Basic Data visualisations

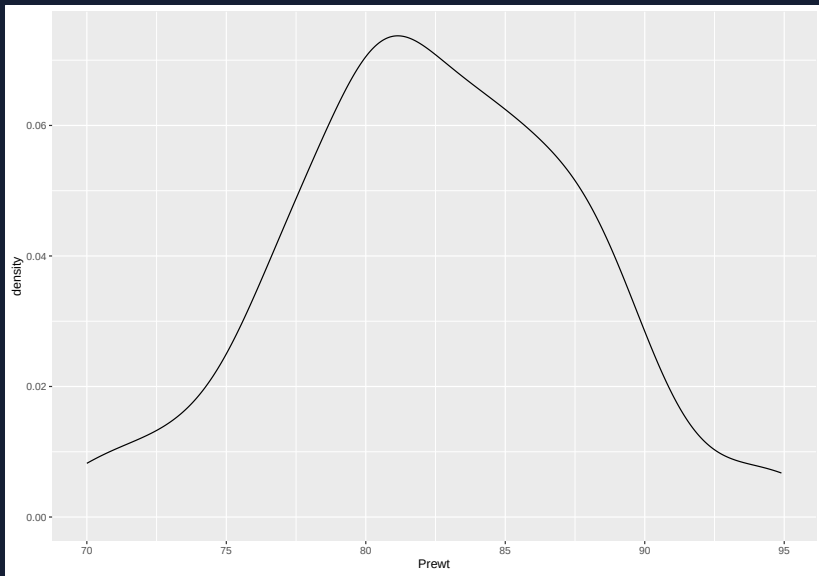
Histograms



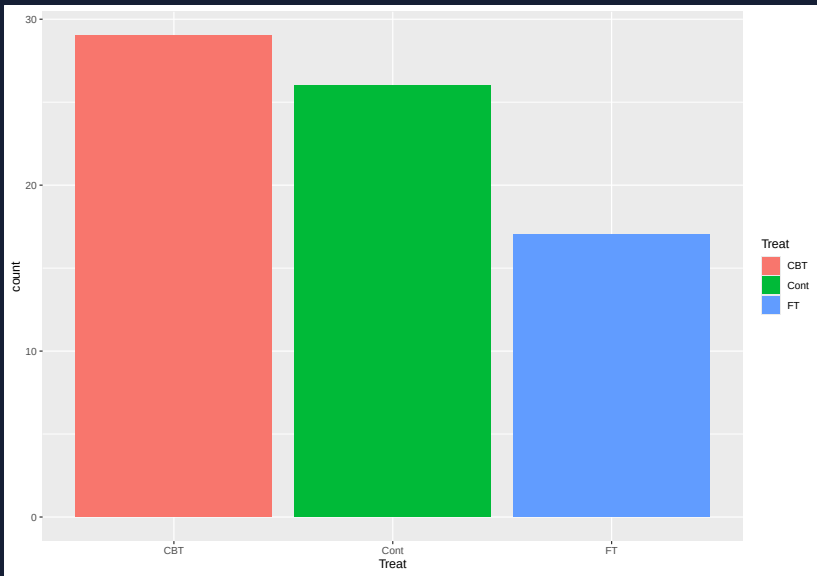
Frequency Polygons



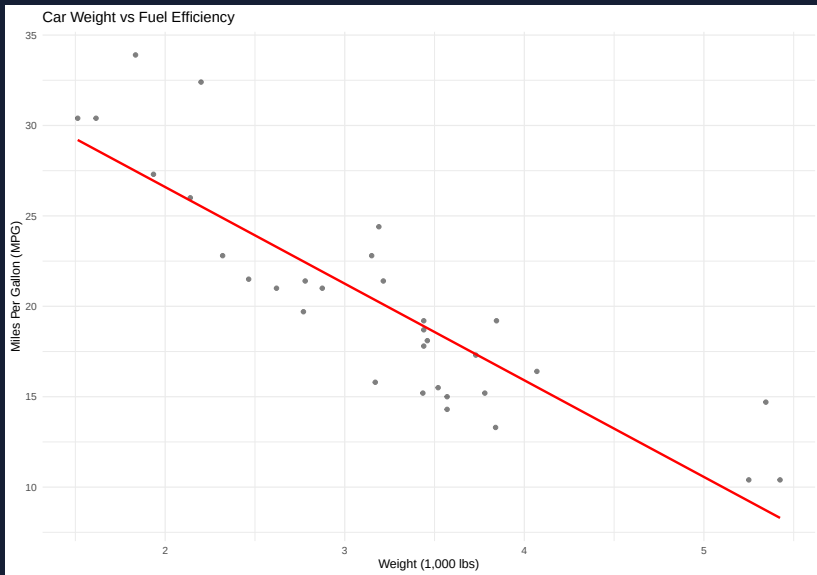
Density Plots



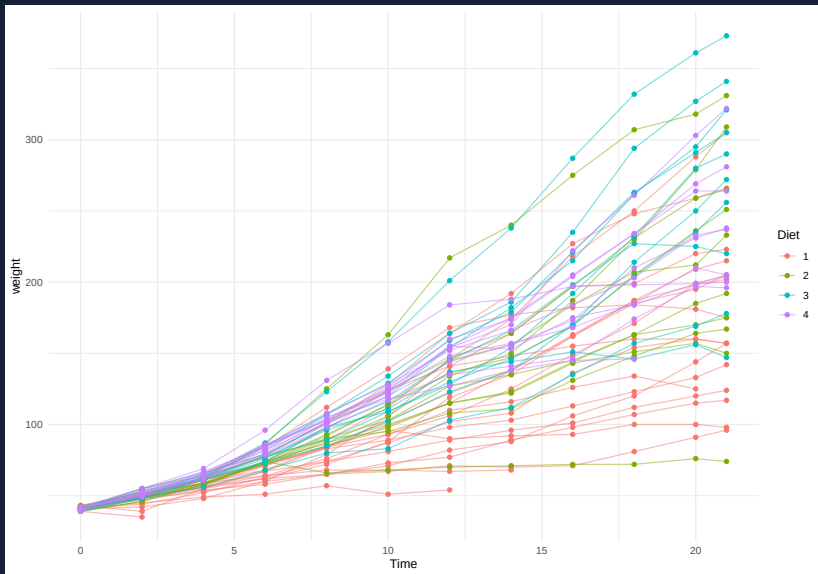
Bar Plots



Scatter Plot



Growth Curve Plots



Facet Plots

```
## Grouped Data: weight ~ Time | Chick
```

```
##      weight Time Chick Diet
```

```
## 1         42    0     1    1
```

```
## 2         51    2     1    1
```

```
## 3         59    4     1    1
```

```
## 4         64    6     1    1
```

```
## 5         76    8     1    1
```

```
## 6         93   10     1    1
```

```
## 7        106   12     1    1
```

```
## 8        125   14     1    1
```

```
## 9        149   16     1    1
```

```
## 10       171   18     1    1
```

```
## 11       199   20     1    1
```

```
## 12       205   21     1    1
```

```
## 13        40    0     2    1
```

```
## 14        49    2     2    1
```

```
## 15        58    4     2    1
```

```
## 16        72    6     2    1
```

```
## 17        84    8     2    1
```

```
## 18       103   10     2    1
```

```
## 19       122   12     2    1
```

```
## 20       138   14     2    1
```

```
## 21       162   16     2    1
```

```
## 22       187   18     2    1
```

```
## 23       209   20     2    1
```

```
## 24       215   21     2    1
```

Psychology Students

For any psychology Students who wish to learn more about R and RStudio then do book a session with us on the Booking System.